

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/233086471>

A Prototype Feeding–Growth Table for White Sturgeon

Article in *Journal of Applied Aquaculture* · January 1995

DOI: 10.1300/J028v05n04_03

CITATIONS

30

READS

548

2 authors, including:



Silas S. O. Hung

University of California, Davis

120 PUBLICATIONS 4,922 CITATIONS

SEE PROFILE

Some of the authors of this publication are also working on these related projects:



Ecological Performance of Fishes in an Ever-changing Estuary: The Effects of Nutritional Status on Environmental Stress Tolerance in Sturgeon [View project](#)

A Prototype Feeding-Growth Table for White Sturgeon

Yibo Cui
Silas S. O. Hung

ABSTRACT. Based upon a series of feeding experiments, an empirical model was developed to predict the growth rate of white sturgeon, *Acipenser transmontanus*, from feeding rate, water temperature, and body weight. A feeding model was also developed to predict optimum feeding rate from body weight and water temperature. Validation of the growth model against independent data showed that the model predictions agreed well with observed growth. A prototype feeding-growth table was established for white sturgeon between 50-1,000 g at 10-26°C. [Article copies available from The Haworth Document Delivery Service: 1-800-342-9678.]

INTRODUCTION

White sturgeon is a relatively new species for aquaculture (Conte et al. 1987), and currently there is no published feeding-growth table for this species. Such a table is needed for use by sturgeon farmers, for design of feeding levels in research, and for evaluating the growth performance under both farmed and experimental conditions. The most important factors affecting the growth of fish include feeding rate, water temperature, and body weight (Brett and Groves 1979). Establishment of a feeding-growth table for sturgeon culture requires knowledge on the quantitative effects of these factors on the growth rate of this species.

Yibo Cui, State Key Laboratory of Freshwater Ecology and Biotechnology, Institute of Hydrobiology, Academia Sinica, Wuhan 430072, Peoples Republic of China.

Silas S. O. Hung, Department of Animal Science, University of California, Davis, CA 95616-8521.

Address correspondence to Silas S. O. Hung.

Eight feeding experiments were conducted at the University of California, Davis (UCD) to investigate the effect of feeding rate on the growth of various sizes of sturgeon kept at different temperatures. Most of the results from the experiments have been published (Hung and Lutes 1987, Hung et al. 1989, 1993, 1995). In this paper, empirical feeding and growth models were established, as well as a prototype feeding-growth table based on the models.

MATERIALS AND METHODS

Model Development

The data for the development of the models come from previous experiments on the effect of feeding rate and water temperature on the growth of white sturgeon (Table 1). Details of the experimental procedures were described previously (Hung and Lutes 1987, Hung et al. 1989, 1993, 1995), and only a brief account is given here. Each experiment was conducted at a constant water temperature. Sturgeon were raised in the laboratory from yolk-sac larvae produced at the sturgeon hatchery at UCD or by local fish farmers. Groups of fish were raised in circular, centrally-drained flow-through tanks, and each tank was randomly assigned to one of several feeding rates fed in triplicates. The tanks were made of fiberglass. Depending on experiment and fish size, the water volume per tank ranged 90-600 L; water turnover rate ranged 1-3 times per hour; and stocking density ranged 25-30 fish per tank. Experiment #8 was conducted in farm-culture systems, which had a water volume of 34,600 L, with a water turnover time of every 3 hours and stocking density of 18.5-36.4 g/L. The diet used in most experiments was a pelleted sturgeon purified diet, which contained 10% moisture, 43% crude protein, 16% crude lipid, and 4% ash (Hung et al. 1987). This diet has been used to raise our hatchery stock for more than 2 years with good growth. In experiment #8 on the largest size class (873 g), a prototype commercial white sturgeon feed was used (Hung et al. 1995). The daily ration was dispensed into the tanks using automatic feeders over a 24-hour period. The experimental duration was 8 weeks, with the exception of one which lasted 6 weeks. The fish were weighed once every 2 weeks. As the fish were not fed on the day they were weighed, the exact length of each growth period was taken as 13 days.

The growth model predicting growth rate from feeding rate, water temperature, and fish weight was developed using a forward selection, step-wise regression technique (Draper and Smith 1981). Calculations were made based on data from each 2-week period. The dependent vari-

TABLE 1. Brief descriptions of the conditions of the feeding experiments. Sources are: 1. Hung and Lutes (1987); 2. Hung et al. (1989); 3. Hung et al. (1993); 4. Hung et al. (1995); 5. unpublished data.

Exp. No.	Water temp. (°C)	Fish weight (g)		Feeding rate (% B.W./day)	Source
		Mean	Range		
1	20	438	310-625	0.5-4.0	5
2	20	50	27-94	2.0-3.5	1
3	18	150	111-195	0.5-2.0	5
4	11	150	111-195	0.5-2.0	5
5	23	72	35-148	0.5-2.0	3
6	26	72	35-148	0.5-2.0	3
7	18	348	250-446	0.5-2.0	2
8	22	873	590-1,186	0.5-1.7	4

able, growth rate (G : %/body weight/day), was calculated as: $G = 100 \times [(W_t - W_0)/W/t]$, where W_t is the final weight, W_0 is the initial fish weight, W is the mean weight $[(W_t + W_0)/2]$, and t is the length of each culture period (13 days). The predictor variables are: $\ln(F + 0.1)$, T , T^2 , $\ln W$, and their interactions, $\ln(F + 0.1) \times T$, $\ln(F + 0.1) \times T^2$, $\ln(F + 0.1) \times \ln W$, $T \times \ln W$, $T^2 \ln W$, $\ln(F + 0.1) \times T \times \ln W$ and $\ln(F + 0.1) \times T^2 \times \ln W$, where $\ln$ is natural logarithm, F is feeding rate (% body weight/day), T is water temperature ($^{\circ}\text{C}$), and W is body weight of fish (g). In the feeding experiments, uneaten feed was not quantified. Therefore, feeding rate represents the rate at which feed was provided to the fish instead of rate of feed intake.

Only eight sets of data are available on the optimum feeding rate of sturgeon. Optimum feeding rate is usually defined as the rate at which the ratio of growth to feed intake is maximal (Brett and Groves 1979). However, in the present paper, in accordance with our previous publications (Hung and Lutes 1987, Hung et al. 1989, 1993), optimum feeding rate is defined as the rate above which growth no longer increases with increased feeding rate. The optimum feeding rate in each experiment was estimated using analysis of variance and multiple range tests and was the minimum rate which produced growth not significantly different from that of fish fed the highest rate. The following model was fitted to the data, using multiple regression technique, to develop the empirical model for optimum feeding rate: $\ln F_{\text{opt}} = a + b_1(\ln W) + b_2T + b_3T^2$, where F_{opt} is optimum feeding rate (%body weight/day), W is fish weight (g) (mean weight of fish in the experiment), and T is water temperature ($^{\circ}\text{C}$).

Model Validation

Independent data from nine published or unpublished experiments conducted in our laboratory were used for the validation of the growth model. These experiments were conducted to study various aspects of nutrition and feeding of white sturgeon. The experimental procedures and facilities were similar to those for the experiments used for model development, and the details can be found in the published papers (see Table 2). A summary of the experimental conditions is provided in Table 2.

Only data for the control group or group(s) with the highest growth rates were used. Data used were the average values for fish reared in individual tanks. The developed growth model was used to predict final body weight of fish based on the initial body weight, water temperature, feeding rate, and experimental duration. The predicted final weight was then compared with the observed weight.

TABLE 2. Brief descriptions of the conditions in the validation experiments. Diet characters are: SPD: sturgeon purified diet (Hung et al. 1987); SC, Silver Cup¹, a commercial nonpurified trout feed (Murray Elevators, Murray, Utah). Sources are: 1. Hung et al. (1987); 2. Hung and Lutes (1988); 3. Moore et al. (1988); 4. Hung (1989); 5. Hung et al. (1989); 6. Stuart and Hung (1989); 7. unpublished data.

Exp. No.	Water temp. (°C)	Initial weight (g)	Feeding rate (%B.W./day)	Duration (days)	Diet characters	N	Source
1	19.4	16.4-24.5	4	56	SPD	3	1
2	19.7	10.8-11.7	2.5	56	SPD	3	2
3	19.8	141-149	1.5	56	SPD	9	3
4	18.8	31.7-35.2	2	42	SPD	3	4
5	19.6	46.2-49.8	2	56	SPD, with 27% maltose	3	5
6	19	15.3-16.4	2	56	SPD	3	6
7	18.5	8.3-8.8	4	56	SPD, with 30% carbohydrate	3	7
8	18.5	7.8-9.4	3.23-3.38	56	SC #2 and #3 semi-moist	4	7
9	20	363	1.5	70	SC #4, dry	1	7
Pooled	18.5-20	7.8-363	1.5-4	56-70		32	

¹Use of trade or manufacturer's name does not imply endorsement.

RESULTS

The growth model can be expressed using the equation:

$$G = -4.43 + 0.566T - 0.0151T^2 + 0.00849\ln(F + 0.1) \times T^2 - 0.00122\ln(F + 0.1) \times T^2 \times \ln W$$

(N = 454, $r^2 = 75.3\%$) Equation 1

The feeding model can be expressed using the equation:

$$\ln F_{\text{opt}} = -2.88 - 0.25\ln W + 0.400T - 0.0077T^2$$

(N = 8, $r^2 = 92.6\%$) Equation 2

The predicted final weight of the fish in the validation experiments was highly correlated with the observed final weight (Figure 1). A linear regression equation was developed relating predicted final weight (W_p) to the observed weight (W_o):

$$W_p = -0.342 + 0.972W_o \quad (N = 32, r^2 = 99.3\%).$$

The model had a high r^2 . The intercept was not significantly different from 0 and the slope was not significantly different from 1 ($P > 0.05$), suggesting that the predicted values agreed well with the observed values.

A prototype feeding-growth table was calculated (Table 3). Suggested feeding rate was calculated from equation 2, and the corresponding growth rate was calculated from equation 1. Calculations were not made for fish larger than 1 kg because fish in both the feeding experiments and validation experiments were smaller than 1.2 kg.

DISCUSSION

Empirical growth models developed by step-wise regression usually showed a good fit to the data. This has been demonstrated by the models developed for the three-spined stickleback, *Gasterosteus aculeatus* (Allen and Wootton 1982), black rockfish, *Sebastes melanops* (Boehlert and Yoklavich 1983), and European minnow, *Phoxinus phoxinus* (Cui and Wootton 1988). Such models, however, are difficult to interpret biologically, and extrapolation of the model outside the experimental conditions may lead to unrealistic predictions. In the present study, the empirical growth model for the white sturgeon fitted the data well and provided accurate

The second limitation is that the maximum weight of fish in both of the feeding and validation experiments was below 1.2 kg. Because of the empirical nature of the models, they should not be extrapolated to fish larger than 1.2 kg. We were not able to test larger sturgeon in the facilities available in our laboratory. Data on the growth-feeding relationship in larger sturgeon are urgently needed.

The third limitation is that the model for optimum feeding rate (equation 2) was derived based only on eight sets of data and was not validated against independent data. More data on the optimum feeding rate of the sturgeon under various combinations of fish size, diet type, and temperature are needed.

Despite these limitations, the feeding-growth table (Table 3) appears reasonable. At a given weight, feeding rate and growth rate generally increase with increased temperature, and at a given temperature, they generally decrease with increased body weight. Some irregularities occurred at lower temperatures, and this was probably due to only one experiment being conducted at temperatures below 18°C. The table shows that fish would lose weight at 10°C, and this was in agreement with the results of the experiment at 11°C, in which most fish did not grow (unpublished data). The table suggests that optimum water temperature for growth (the temperature at which growth rate is highest) tends to decrease with increasing fish size, but it is not clear whether this pattern is realistic or not, because insufficient data are available on the combined effects of temperature and body size on sturgeon growth. The models and the feeding-growth table can be used as an initial guide for feeding rates under farmed and experimental situations and to provide a standard for evaluating sturgeon growth. However, because of the preliminary and empirical nature of the models and because many factors, including water quality, feed quality, and feeding strategy may affect the growth-feeding relationship in fish, caution is advised when applying the models, especially under farm-production situations, and the adjustments of feeding rates need to be made according to growth performance. The models and feeding-growth table should not be used on fish larger than 1 kg.

ACKNOWLEDGMENTS

Y. Cui was supported by the Chinese Academy of Sciences for his visit to Davis. The authors thank Drs. Jim Fadel and Fred Conte for their comments on the manuscript.

REFERENCES

- Allen, J.R.M., and R.J. Wootton. 1982. The effect of ration and temperature on the growth of the three-spined stickleback, *Gasterosteus aculeatus* L. *Journal of Fish Biology* 20:409-422.
- Boehlert, G.W., and M.M. Yoklavich. 1983. Effects of temperature, ration, and fish size on growth of juvenile black rockfish. *Environmental Biology of Fishes* 8:17-28.
- Brett, J.R., and T.D.D. Groves. 1979. Physiological energetics. Pages 279-352 in W.S. Hoar, D.J. Randall, and J.R. Brett, eds. *Fish Physiology*, Vol. 8. Academic Press, London.
- Conte, F.S., S.I. Doroshov, P.B. Lutes, and E.M. Strange. 1987. Sturgeon Hatchery Manual for the White Sturgeon (*Acipenser transmontanus* Richardson), with Application to Other North American Acipenseridae. Publication 3322, Division of Agriculture and Natural Resources, University of California, Oakland, CA.
- Cui, Y., and R.J. Wootton. 1988. Bioenergetics of growth of a cyprinid, *Phoxinus phoxinus* (L.): the effect of ration and temperature on growth rate and efficiency. *Journal of Fish Biology* 33:763-773.
- Draper, N.R., and H. Smith. 1981. *Applied Regression Analysis*. John Wiley, New York, NY.
- Hung, S.S.O. 1989. Choline requirement of hatchery-produced juvenile white sturgeon (*Acipenser transmontanus*). *Aquaculture* 78:183-194.
- Hung, S.S.O., and P.B. Lutes. 1987. Optimum feeding rate of hatchery-produced juvenile white sturgeon (*Acipenser transmontanus*) at 20°C. *Aquaculture* 65:307-317.
- Hung, S.S.O., and P.B. Lutes. 1988. A preliminary study on the non-essentiality of lecithin for hatchery-produced juvenile white sturgeon (*Acipenser transmontanus*). *Aquaculture* 68:353-360.
- Hung, S.S.O., F.S. Conte, and P.B. Lutes. 1995. Optimum feeding rate of white sturgeon, *Acipenser transmontanus*, yearlings under commercial production conditions. *Journal of Applied Aquaculture* 5(1):45-51.
- Hung, S.S.O., B.J. Moore, C.E. Bordner, and F.S. Conte. 1987. Growth of juvenile white sturgeon (*Acipenser transmontanus*) fed different purified diets. *Journal of Nutrition* 117:328-334.
- Hung, S.S.O., F.K. Fynn-Aikins, P.B. Lutes, and R.P. Xu. 1989. Ability of juvenile white sturgeon (*Acipenser transmontanus*) to utilize different carbohydrate sources. *Journal of Nutrition* 119:727-733.
- Hung, S.S.O., P.B. Lutes, F.S. Conte, and T. Storebakken. 1989. Growth and feed efficiency of white sturgeon (*Acipenser transmontanus*) sub-yearlings at different feeding rates. *Aquaculture* 80:147-153.
- Hung, S.S.O., P.B. Lutes, A.A. Shqueir, and F.S. Conte. 1993. Effect of feeding rate and water temperature on growth of juvenile white sturgeon (*Acipenser transmontanus*). *Aquaculture* 115:297-303.
- Moore, B.J., S.S.O. Hung, and J.F. Medrano. 1988. Protein requirement of hatchery-produced juvenile white sturgeon (*Acipenser transmontanus*). *Aquaculture* 71:235-245.
- Stuart, J.S., and S.S.O. Hung. 1989. Growth of juvenile white sturgeon (*Acipenser transmontanus*) fed different proteins. *Aquaculture* 76:303-316.